**Computer Vision Lab Sheet**

**Experiment 3**

**Name –** Vishavjeet Baraily **CUID** – CU24250187

**Branch** – B. Tech CSE **Year** – 3rd (Vth sem) Sec - B

**Roll No**. – 68

**# Step 1**. Import the required libraries and load a grayscale or color image for processing.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
print("Step 1: Upload an image file for processing...")
uploaded = files.upload()
file_name = list(uploaded.keys())[0]
# Load image in grayscale[cite: 1]
img = cv2.imread(file_name, cv2.IMREAD_GRAYSCALE)
if img is None:
    raise ValueError("Error loading image. Please upload a valid image file.")
```

**# Step 2**. Display the original image and analyze its visual characteristics.

```
print("\nStep 2: Original Image Characteristics")
print(f"- Image Shape (Height, Width): {img.shape}")
print(f"- Data Type: {img.dtype}")
print(f"- Intensity Range: Min = {img.min()}, Max = {img.max()}")
```

**# Step 3**. Apply Gaussian Blur to reduce image noise while preserving important structures.

```
gaussian_blur = cv2.GaussianBlur(img, (5, 5), 0)
```

**# Step 4.** Apply Median Filtering to remove impulse (salt-and-pepper) noise from the image.

```
median_blur = cv2.medianBlur(img, 5)
```

**# Step 5.** Apply Average (Mean) Filtering and compare its smoothing performance with Gaussian filtering.

```
average_blur = cv2.blur(img, (5, 5))
```

**# Step 6**. Implement Laplacian Filtering to enhance image edges and fine details.

```
laplacian = cv2.Laplacian(img, cv2.CV_64F)

laplacian_out = np.uint8(np.absolute(laplacian))
```

**# Step 7.** Apply Sobel Edge Detection in both horizontal and vertical directions to detect image gradients.

```
sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3) # Horizontal

sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3) # Vertical

sobel_combined = cv2.magnitude(sobel_x, sobel_y)      # Combined Magnitude

sobel_out = np.uint8(np.absolute(sobel_combined))
```

**# Step 8.** Compare the outputs of different low-pass and high-pass filters in terms of noise reduction, edge preservation, and computational performance.

**# Step 9.** Display the original and filtered images along with suitable observations highlighting the strengths and limitations of each filtering technique.

```
titles = [
    '1. Original Image',
    '2. Average Filter (LPF)',
    '3. Gaussian Blur (LPF)',
    '4. Median Filter (LPF)',
    '5. Laplacian Filter (HPF)',
    '6. Sobel Filter (HPF)'
]
images = [img, average_blur, gaussian_blur, median_blur, laplacian_out, sobel_out]


plt.figure(figsize=(16, 10))
for i in range(6):
    plt.subplot(2, 3, i + 1)
    plt.imshow(images[i], cmap='gray')
```

```
    plt.title(titles[i], fontsize=11, fontweight='bold')
    plt.axis('off')

plt.tight_layout()
plt.show()
```

**# Step 10.** Interpret the results and discuss the applications of spatial filtering in real-world computer vision systems.

```
print("\n" + "="*70)
print("OBSERVATIONS & ANALYSIS (Steps 8, 9 & 10)")
print("="*70)

print("""
```

1. Low-Pass Filters (LPF):
   - Average Filter: Blurs the image uniformly by taking the mean of pixels, causing notable loss of edge sharp detail.
   - Gaussian Blur: Smoothes the image more naturally using weighted Gaussian distribution, preserving edges significantly better than Average filter.
   - Median Filter: Non-linear filter that removes high-contrast impulse/salt-and-pepper noise cleanly while preserving crisp boundary lines.

2. High-Pass Filters (HPF):
   - Laplacian Filter: Second-order operator that detects rapid intensity changes equally in all directions, highlighting fine lines and rapid transitions.
   - Sobel Operator: First-order derivative operator that effectively isolates structural directional gradients (edges) in both horizontal and vertical directions.

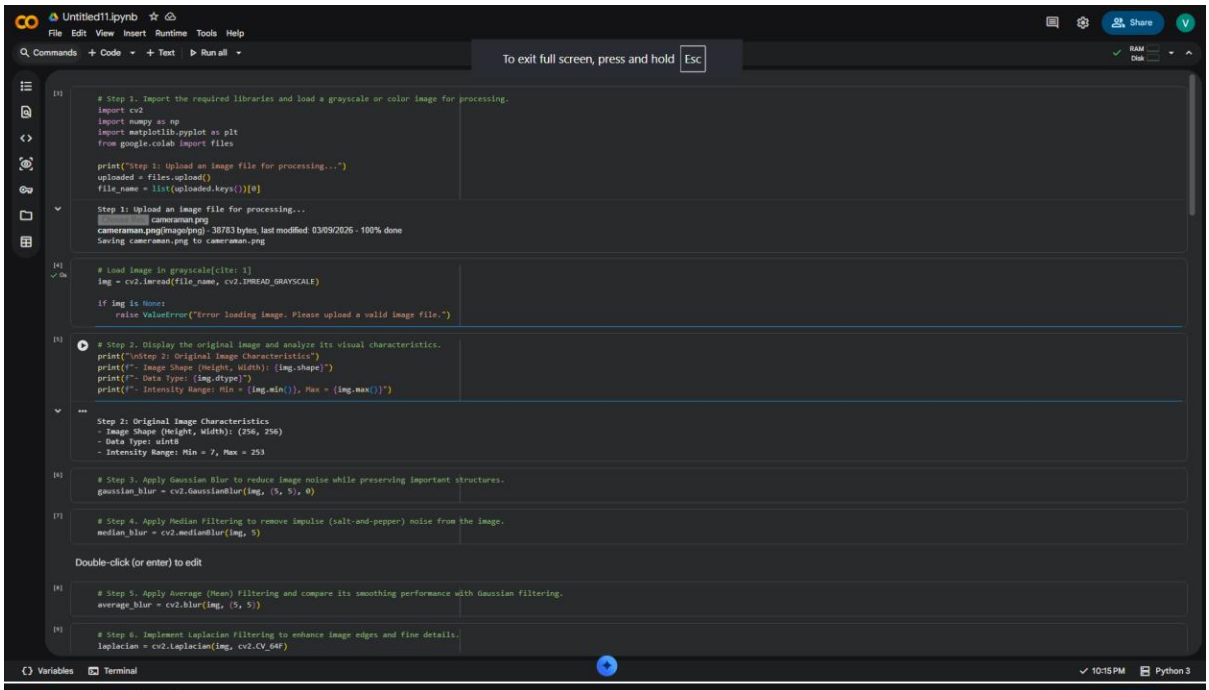
3. Real-World Applications:
   - Medical Imaging: Denoising X-ray, CT, or MRI scans without blurring structural boundaries.
   - Autonomous Vehicles: Edge and line detection (via Sobel) for real-time lane tracking.

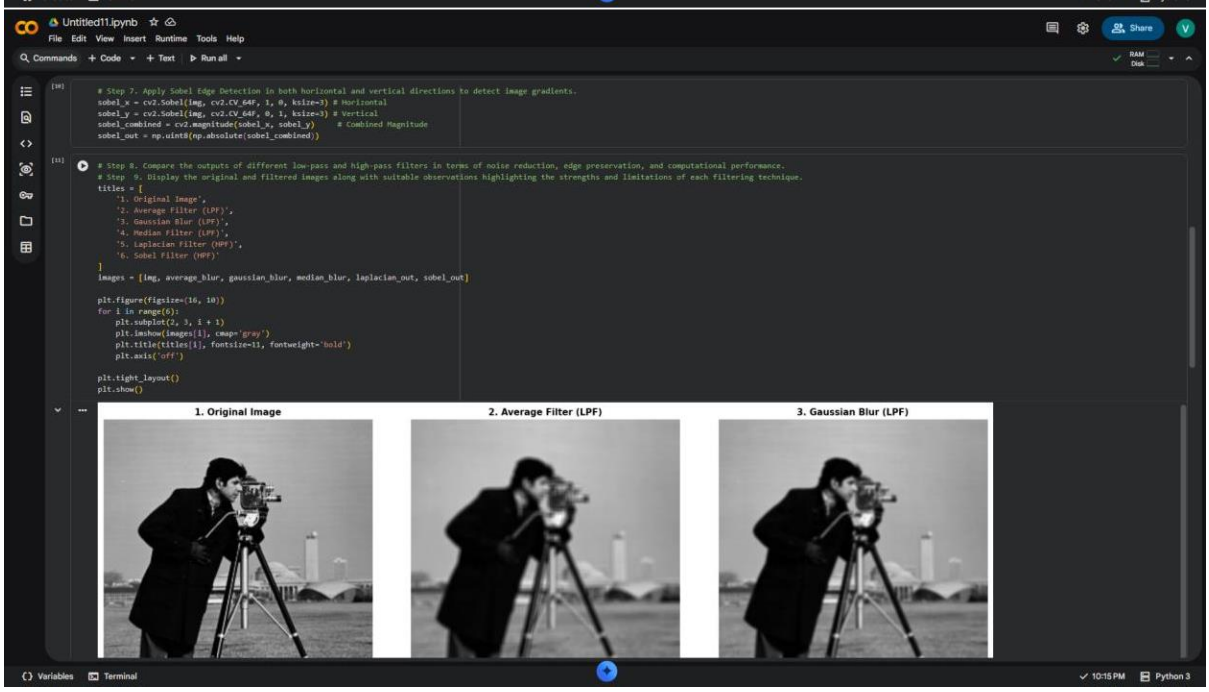
- Satellite Imaging: Sharpening terrain feature definitions for geographic analysis.

- Surveillance: Preprocessing low-light security footage to reduce sensor noise.

""")

**Outputs** –



1. Original Image



2. Average Filter (LPF)



3. Gaussian Blur (LPF)



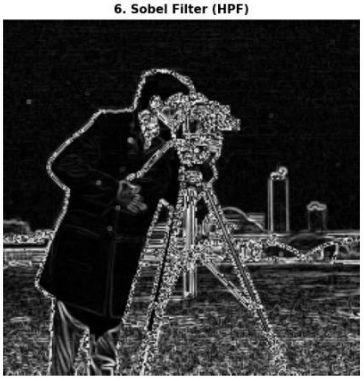
4. Median Filter (LPF)



5. Laplacian Filter (HPF)



6. Sobel Filter (HPF)



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```
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...

======================================================================
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======================================================================

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```

Variables Terminal

10:15 PM Python 3

**Questions**

1. What is spatial filtering? How is it used in digital image processing?

Spatial filtering passes a small matrix (kernel) directly over pixel neighborhoods to calculate a new pixel value. It modifies image characteristics in the spatial domain for noise removal, smoothing, and edge enhancement.

2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.

- Low-Pass Filters (LPF): Pass low frequencies (smooth areas) and attenuate high frequencies (edges/noise). Example: Gaussian Blur, Average Filter.
- High-Pass Filters (HPF): Pass high frequencies and suppress low frequencies to highlight fine details and boundaries. Example: Laplacian Filter, Sobel Operator.

3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications.

Average

- Working Principle - Replaces target pixel with the linear mean of its neighborhood.
- Application - Basic noise reduction, general blurring.

Gaussian

- Working Principle - Uses a weighted average based on a 2D Gaussian distribution.
- Application - Preserving natural transitions while reducing Gaussian noise.

Median

- Working Principle - Replaces target pixel with the statistical median of neighborhood values.
- Application - Removing impulsive, high-contrast salt-and-pepper noise.

4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?

Salt-and-pepper noise creates extreme pixel values (0 or 255). Non-linear median filtering ignores extreme outliers when sorting, replacing the noisy pixel with a representative neighborhood value without smearing edge details.

5. Explain the role of convolution kernels in spatial filtering.

Kernels are small matrix structures (e.g., $3 \times 3$, $5 \times 5$) containing spatial weights. Through 2D convolution—multiplying kernel weights by underlying pixel values and summing the results—kernels dictate whether the filter acts as a smoother, sharpener, or edge detector.

6. What is the purpose of the Sobel and Laplacian operators in edge detection?

- Sobel Operator: Calculates first-order image gradients to detect directional edges (horizontal and vertical).
- Laplacian Operator: Calculates second-order derivatives to highlight regions of rapid intensity change independently of edge direction.

7. Why are filtering operations considered an essential preprocessing step in computer vision?

Filtering eliminates sensor artifacts and environmental noise while accentuating dominant geometric structures, significantly improving the accuracy of downstream tasks like segmentation, feature extraction, and classification.

8. Discuss the trade-off between image smoothing and edge preservation during filtering.

Linear LPFs blur sharp intensity transitions (edges) alongside noise, resulting in loss of spatial resolution and feature localization. Non-linear LPFs or edge-preserving filters minimize blur but can be computationally expensive.

9. Mention four real-world applications where spatial filtering techniques are widely used.

- Medical Imaging (enhancing X-ray/MRI scan contrast)
- Autonomous Vehicles (detecting lane markers via gradient filters)
- Satellite Remote Sensing (sharpening terrain features)
- Security & Surveillance (denoising low-light security footage)

10. Compare spatial domain filtering with frequency domain filtering in terms of implementation and practical applications.

Spatial Domain Filtering

- Implementation - Directly convolves masks with pixels in the spatial grid.
- Complexity - Simple for small kernel sizes ($3 \times 3, 5 \times 5$).
- Practical applications - Real-time edge detection and quick smoothing.

Frequency Domain Filtering

- Implementation - Applies Fourier Transform (FFT), filters frequencies, applies Inverse FFT.
- Complexity - Efficient for large-scale or global periodic noise removal.
- Practical applications - Removing periodic pattern noise and specialized global frequency manipulation.